Answer Key & Teacher Notes

Session 1 — Substitute and Solve · Session 2 — Working Backward to a Goal

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Substitute and Solve

TALK PROMPTS

- Why should you check an answer against the situation before you trust it?
- How would you check that 216 candles makes sense for 12 hours of work?

PARTNER TASK

Muffin Trays. A bakery bakes 12 muffins on every tray. One partner writes m = 12t and finds the muffins on 14 trays (168); the other estimates (10 × 15 = 150) to check that the answer is reasonable. Switch with Keisha rowing 40 meters per minute for 12 minutes (480).

WATCH FOR

- Adding the input to the rate instead of multiplying (6m with m = 30 gives 36).
- Adding the fee before multiplying ((14 + 25) × 6 instead of 14 × 6 + 25).
- Giving a number without units, or not answering the yes/no question.

SESSION 2 · TEACHER NOTES

Working Backward to a Goal

TALK PROMPTS

- Which is easier for the class to control: the price or the number sold? Why?
- How would your work change if the club raised the ticket price but kept the same goal?

PARTNER TASK

Candle Sale Two Ways. A club needs $540. One partner finds the candles needed at $9 each (60); the other finds the price if the club sells 45 candles ($12). Then each checks the other by multiplying.

WATCH FOR

- Multiplying by the rate instead of dividing (4b = 96 answered as 384).
- Dividing before subtracting the fee (600 ÷ 15 = 40 instead of 510 ÷ 15 = 34).
- Rounding 37.5 tickets down to 37, which does not reach the goal.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.EE.C.9	1	2	1	1	2	2	1	3	2	1	1
S1 · Form B	6.EE.C.9	1	2	1	1	2	2	1	3	2	1	1
S2 · Form A	6.EE.C.9	1	2	1	1	2	3	2	3	2	1	1
S2 · Form B	6.EE.C.9	1	2	1	1	2	3	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

substitute: Put a known number in **place** of a letter.

order of operations: **Multiply** first, then add or subtract.

reasonable: Makes **sense** for the story.

- To find an output, **substitute** the input for the variable.
- With a fee, **multiply** first, then add the fee.

Example: Substitute $h = 5$: $r = 24 \times \mathbf{5}$. Multiply: $r = \mathbf{120}$ rolls. Compare: $120 < 150$. Answer: **No**, 5 hours makes only 120 rolls.

YOU TRY

1. a. 31 b. 108

2. 8 × 5 + 15 = 40 + 15 = $55

EXIT TICKET

1. a. 60 b. 23

2. 14 × 35 = 490 boxes

3. **B** (22)

PRACTICE FORM A

1. a. 72 b. 30 c. 47 d. 73

2. a. y = 18x b. 18 × 12 = 216 candles c. No: 18 × 7 = 126, which is less than 900

3. c = 24t; 24 × 16 = 384 cookies

4. m = 50n; 50 × 9 = 450 meters

5. Yes. 8 × 45 = 360 sandbags, which matches the claim.

6. b = 2.5h; 2.5 × 24 = 60 boards

7. 14 × 6 + 25 = 84 + 25 = $109

8. a. 3 × 15 + 20 = $65 b. 3 × 30 + 20 = $110 c. $110 is not 2 × $65 = $130, because the $20 fee is paid only once.

9. He added 6 + 30 instead of multiplying. 6m means 6 × m, so b = 6 × 30 = 180 buckets.

10. **B** (48)

11. **B** (multiply first, then add the fee)

Self-check word **ANSWER**

PRACTICE FORM B

1. a. 56 b. 32 c. 59 d. 68

2. a. y = 16x b. 16 × 11 = 176 bars c. Yes: 16 × 8 = 128, which is at least 120

3. m = 12t; 12 × 14 = 168 muffins

4. d = 40n; 40 × 12 = 480 meters

5. No. 9 × 30 = 270 boxes, not 290, so the claim does not fit.

6. b = 1.5h; 1.5 × 16 = 24 bikes

7. 12 × 5 + 30 = 60 + 30 = $90

8. a. 4 × 10 + 15 = $55 b. 4 × 20 + 15 = $95 c. $95 is not 2 × $55 = $110, because the $15 fee is paid only once.

9. He added 8 + 20 instead of multiplying. 8m means 8 × m, so j = 8 × 20 = 160 jugs.

10. **B** (63)

11. **B** (multiply 7 by 3)

Self-check word **RESULT**

GUIDED NOTES

work backward: Start with the **goal** and undo the steps.

inverse operation: An action that **undoes** another action.

at least: That number **or more**.

- Put the goal in place of the **output**, then solve for the input.
- With a fee, **subtract** the fee first, then divide by the rate.
- If only whole items make sense, round **up** to reach the goal.

Example: Put in the goal: 100 = **6***n. *Divide: 100 ÷ 6 ≈ 16.7. 16 tickets make $96 (too low); 17** tickets make $102. Answer: The club must sell **17** tickets.

YOU TRY

1. a. x = 48 b. x = 10

2. 60 ÷ 8 = 7.5; 7 vans hold only 56, so 8 vans

EXIT TICKET

1. a. n = 12 b. n = 11

2. 11c = 528; c = 528 ÷ 11 = 48 cars

3. **B** (subtracting 6)

PRACTICE FORM A

1. a. x = 14 b. x = 15 c. x = 11 d. x = 25

2. a. m = 12p b. 660 = 12p, so p = 660 ÷ 12 = 55 pots c. 660 ÷ 44 = $15 each

3. 3n = 210; n = 210 ÷ 3 = 70 pages

4. 7t = 350; t = 350 ÷ 7 = 50 tickets

5. r = 9m; 135 = 9m, so m = 135 ÷ 9 = 15 minutes

6. No: 13 × 25 = $325, which is more than $310. Most: 310 ÷ 25 ≈ 12.4, so 12 tickets.

7. a. m = 15k + 90 b. 600 − 90 = 510; 510 ÷ 15 = 34 tickets

8. a. t = 300 ÷ 8 = 37.5 b. 38 tickets (37 × 8 = $296 is not enough; 38 × 8 = $304)

9. She multiplied 96 × 4. To undo multiplying by 4, divide: b = 96 ÷ 4 = 24 wristbands. Check: 4 × 24 = 96.

10. **B** (13)

11. **C** (8)

Self-check word **INVERSE**

PRACTICE FORM B

1. a. x = 12 b. x = 17 c. x = 13 d. x = 15

2. a. m = 9p b. 540 = 9p, so p = 540 ÷ 9 = 60 candles c. 540 ÷ 45 = $12 each

3. 5n = 185; n = 185 ÷ 5 = 37 laps

4. 6t = 324; t = 324 ÷ 6 = 54 tickets

5. r = 7m; 168 = 7m, so m = 168 ÷ 7 = 24 minutes

6. Yes: 11 × 18 = $198, which is within $200. Most: 200 ÷ 18 ≈ 11.1, so 11 tickets.

7. a. m = 16k + 80 b. 560 − 80 = 480; 480 ÷ 16 = 30 T-shirts

8. a. t = 300 ÷ 9 = 33.3… b. 34 tickets (33 × 9 = $297 is not enough; 34 × 9 = $306)

9. He multiplied 85 × 5. To undo multiplying by 5, divide: s = 85 ÷ 5 = 17 stickers. Check: 5 × 17 = 85.

10. **A** (17)

11. **B** (6)

Self-check word **REVERSE**

SHEET 1 · SUBSTITUTE AND SOLVE

1. a. 19 b. 72 c. 28 d. 42
2. 3, 5, 7, 11, 13; points (0, 3), (1, 5), (2, 7), (4, 11), (5, 13) on a straight line
3. a. 15 × 6 + 20 = 110 gal b. 15 × 10 + 20 = 170 gal
4. 3 hours: 7 × 3 + 5 = $26; 8 hours: 7 × 8 + 5 = $61
5. a. Yes: 5 × 4 + 2 = 22 b. No: 5 × 6 + 2 = 32, not 30
6. He added before multiplying. Multiply first: 4 × 5 = 20, then 20 + 3 = 23. y = 23.

SHEET 2 · WORKING BACKWARD TO A GOAL

1. a. 9 b. 13 c. 9 d. 8
2. 210 − 45 = 165; 165 ÷ 15 = 11 weeks
3. 150 − 30 = 120; 120 ÷ 12 = 10 weeks. Check: 12 × 10 + 30 = 150.
4. a. week 4 b. 33 − 6 = 27; 27 ÷ 3 = 9, so week 9
5. 150 ÷ 12 = 12.5. She cannot buy half a box, and 12 boxes (144) is not enough, so she buys 13 boxes.
6. **B** (6 hours)
7. **C** (x = 13)

October 2026 conceptual additions — teacher key

Session 1 · Substitute and Solve

c=7(4)+12=40; exactly enough, assuming the given model has no other fees.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: c=7(4)+12=40; exactly enough, assuming the given model has no other fees.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Working Backward to a Goal

Need 17: 16 gives $96, 17 gives $102. Can buy at most 16: 16 costs $96, 17 costs $102. Minimum-needed rounds up; maximum-affordable rounds down.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Need 17: 16 gives $96, 17 gives $102. Can buy at most 16: 16 costs $96, 17 costs $102. Minimum-needed rounds up; maximum-affordable rounds down.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.